

Depth and size perception

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Theories of colour vision

1. Trichromatic theory (Young, 1802; von Helmholtz, 1852)

Full colour vision is based on the ability to combine three different wavelengths.

In Trichromats (people with full colour vision), this is achieved by the collaboration of three different types of cones that selectively respond to short, medium, and long wavelength.

2. Opponent process theory (Hering, 1878)

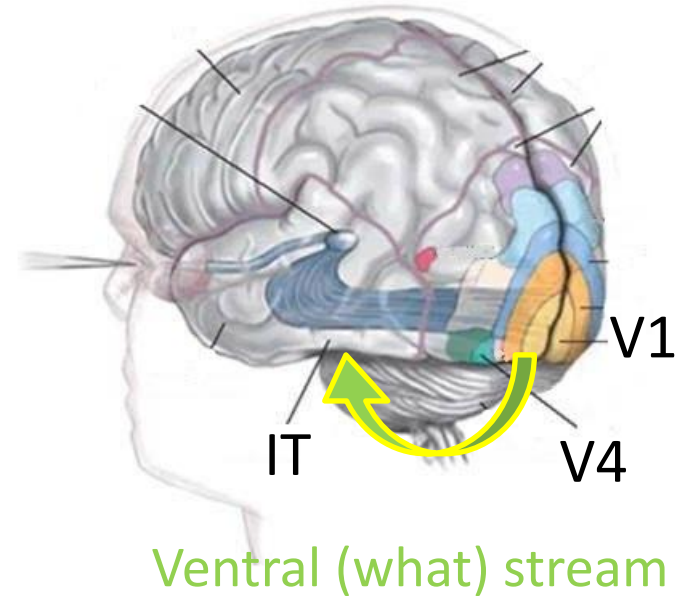
There are three opponent colour channels which respond in opposite ways to different wavelengths:

black/white; red/green; blue/yellow

Theories of colour vision (2)

- Opponent-process theory
 - Physiological evidence

Opponent neurons in V1, V4, and Inferotemporal cortex...

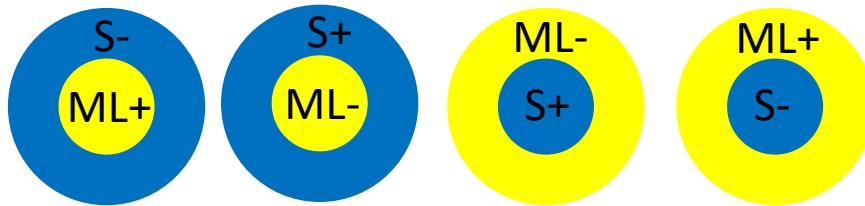


... with **single-opponent** or **double-opponent** receptive fields.

Theories of colour vision (2)

- Opponent-process theory

- **Opponent neurons with single-opponent** receptive fields



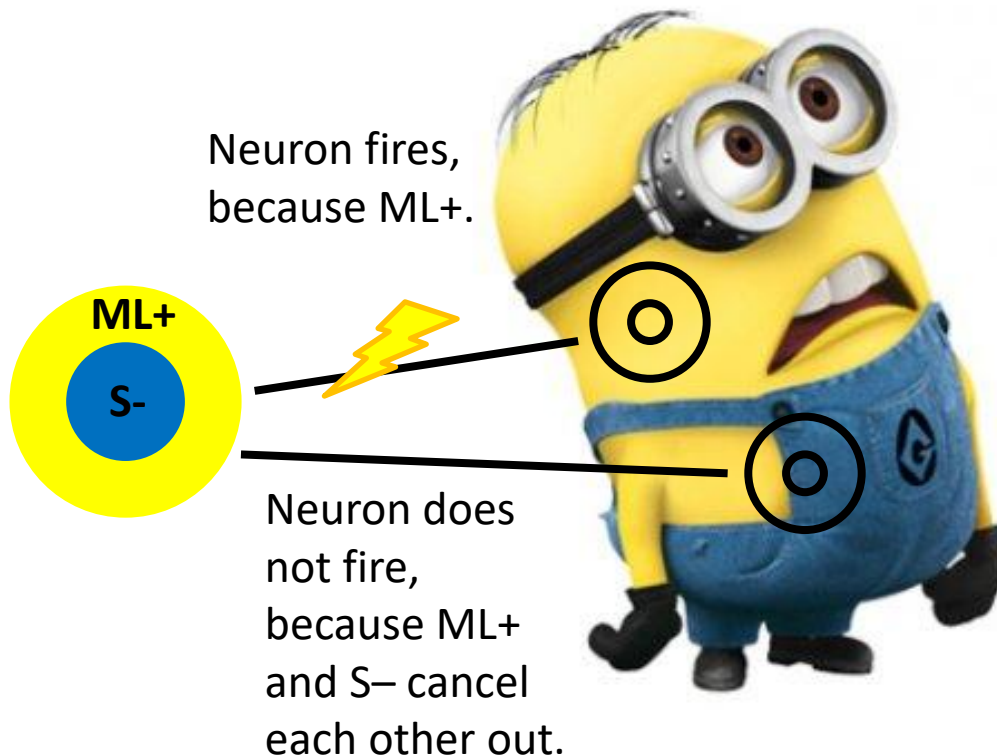
ML = medium & long waves
S = short waves
+ = excitation
- = inhibition

Single-opponent neurons are excited by one colour of the opponent channel and inhibited by the other colour of that channel.

Theories of colour vision (2)

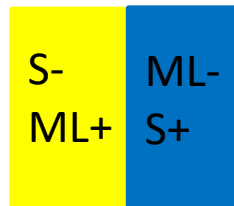
- Opponent-process theory

Neurons with **single-opponent** receptive fields are good at **detecting uniform colour surfaces**.



Theories of colour vision (2)

- Opponent-process theory
 - **Opponent neurons with double-opponent** receptive fields



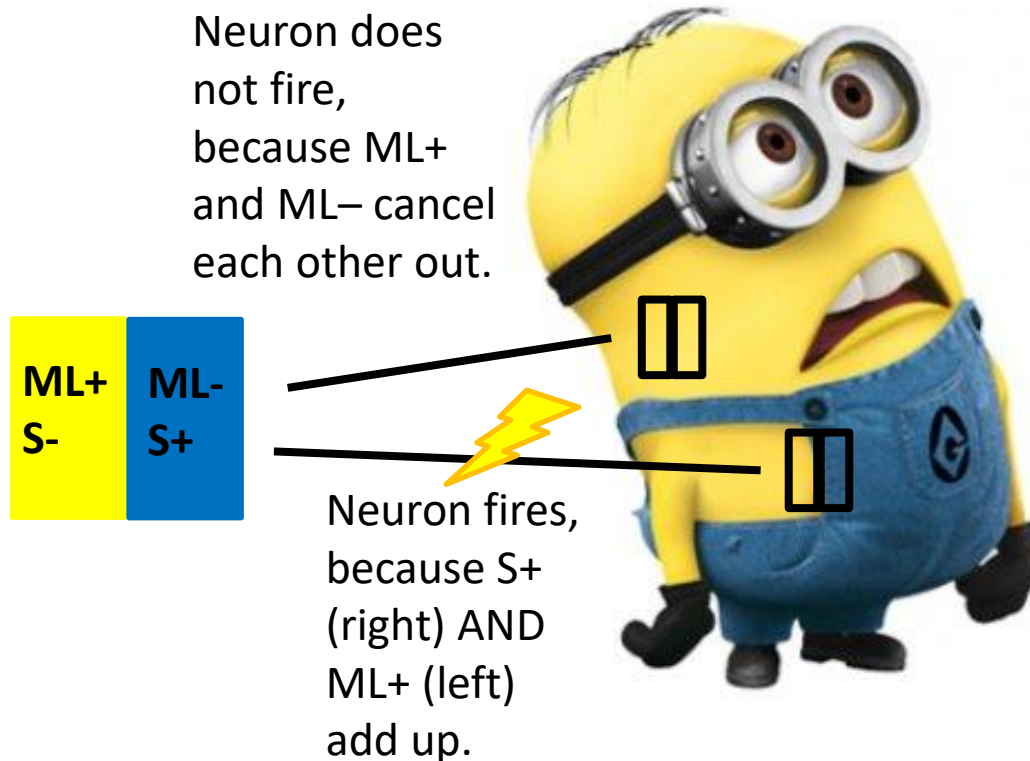
ML = medium & long waves
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Double-opponent neurons are excited and inhibited by both colours of the opponent channel – but on opposite sides of the receptive field.

Theories of colour vision (2)

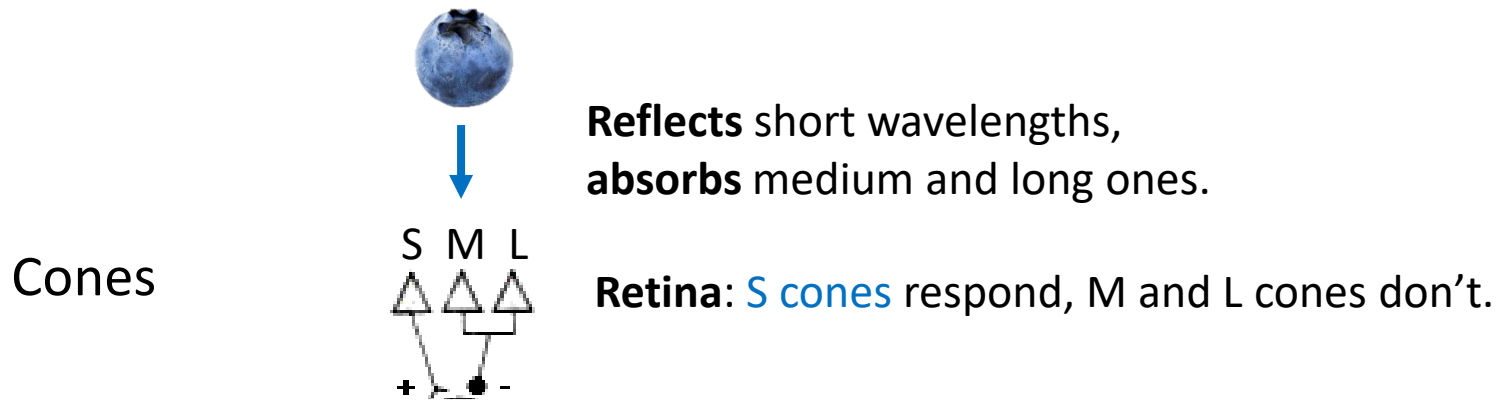
- Opponent-process theory

Neurons with **double-opponent** receptive fields are good at **detecting colour boundaries**.



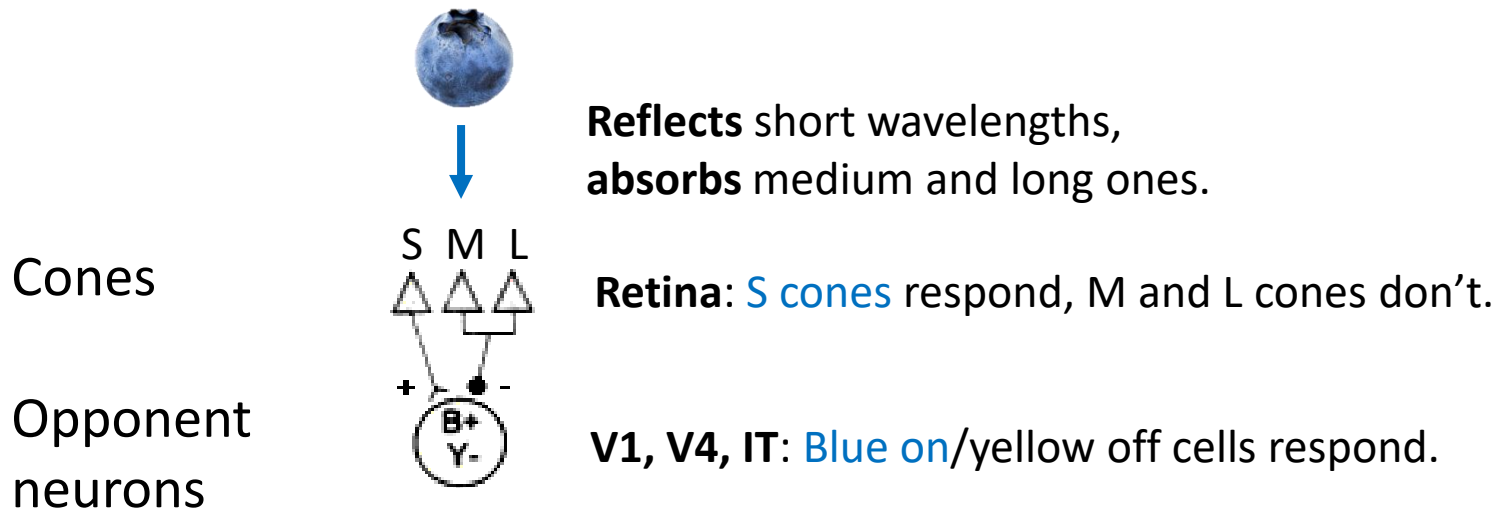
Theories of colour vision – which one is correct?

- Trichromatic theory or opponent-process theory?
 - **BOTH** are correct!



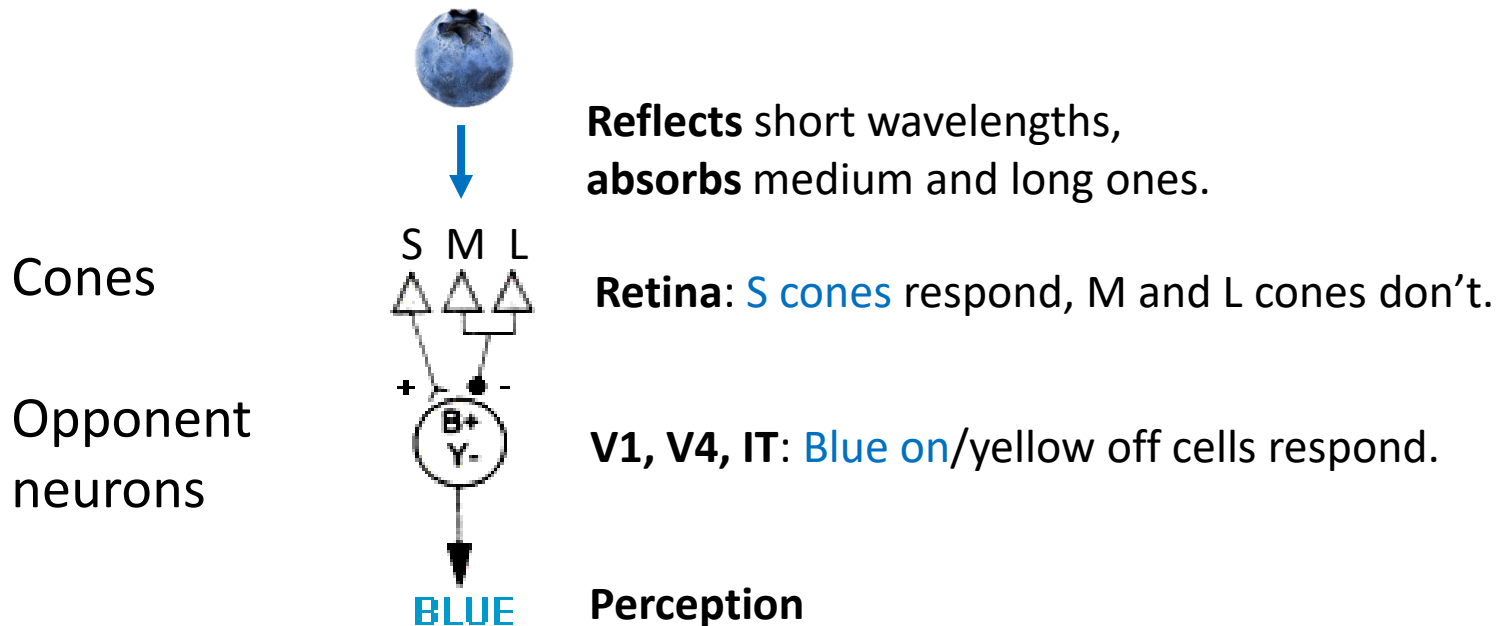
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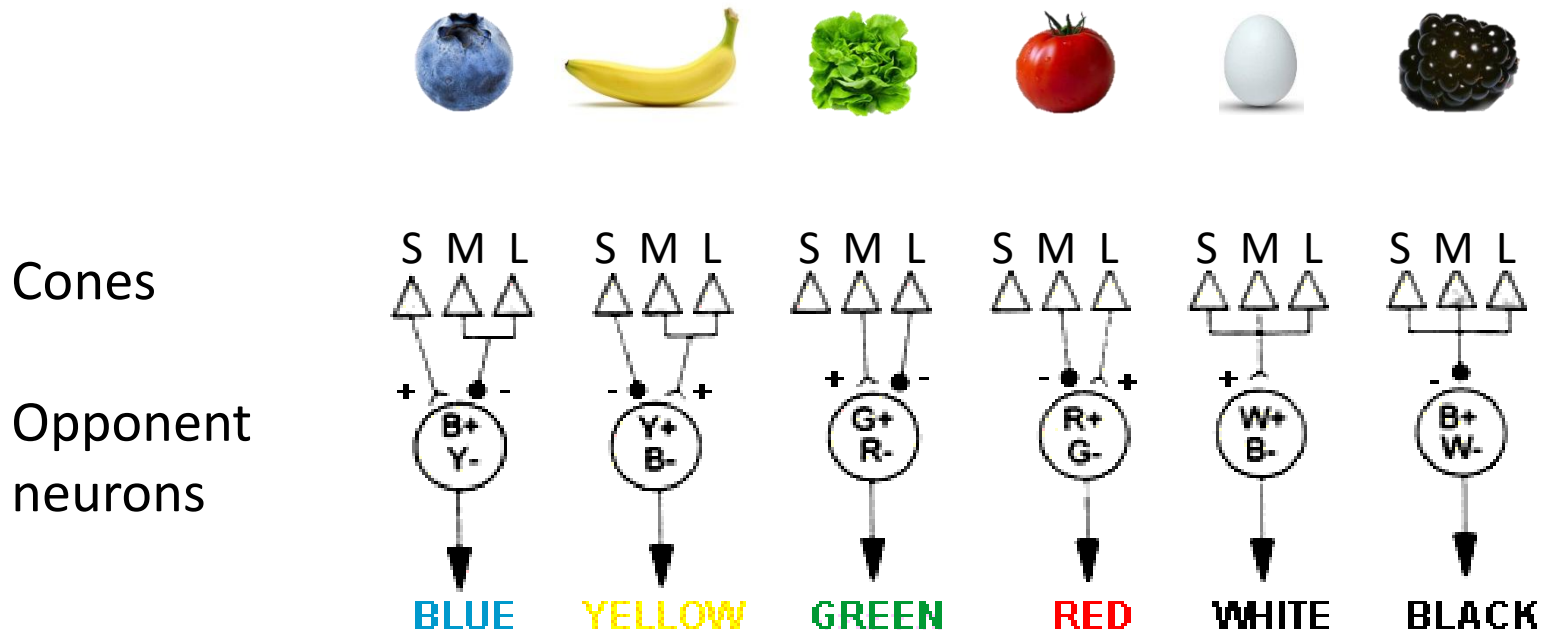
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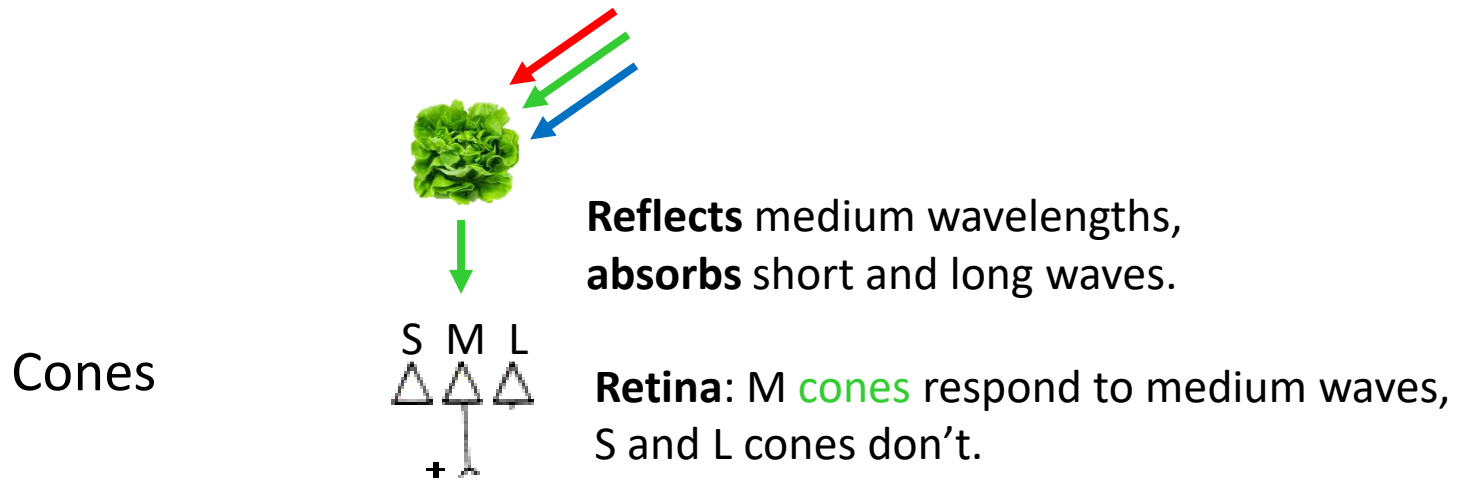
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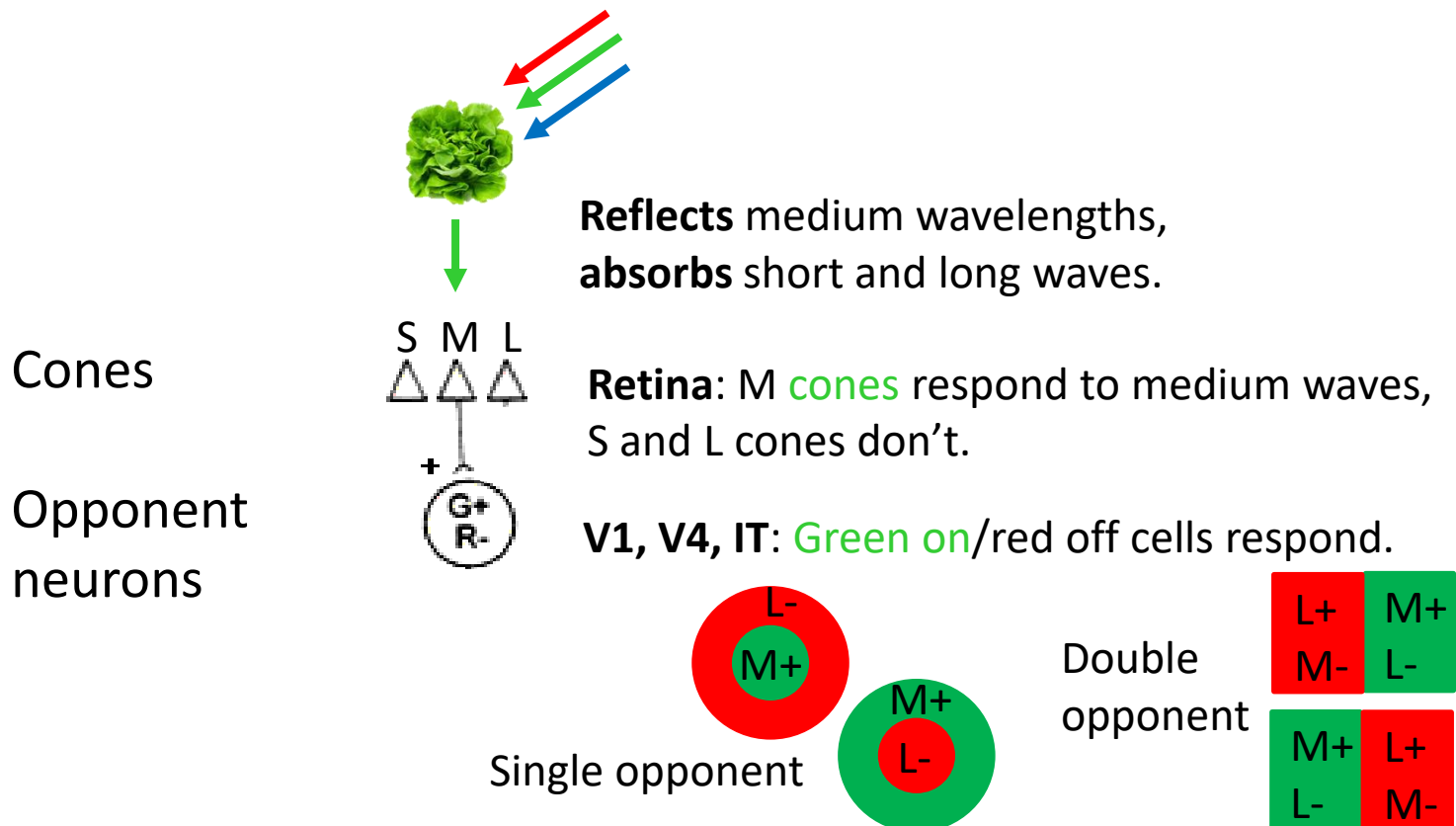
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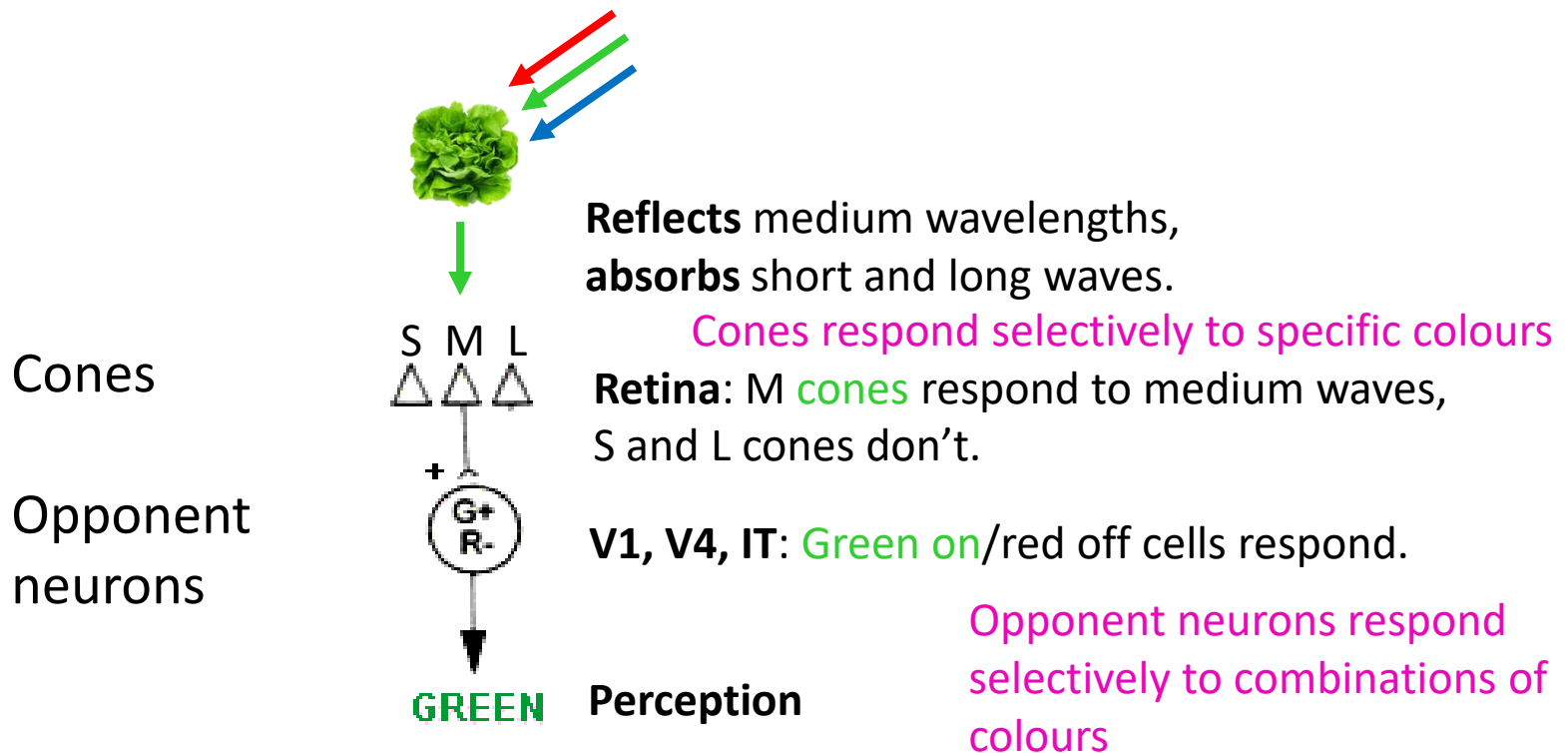
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Theories of colour vision – which one is correct?

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 - **BOTH** are correct!



Last time...

Sensation

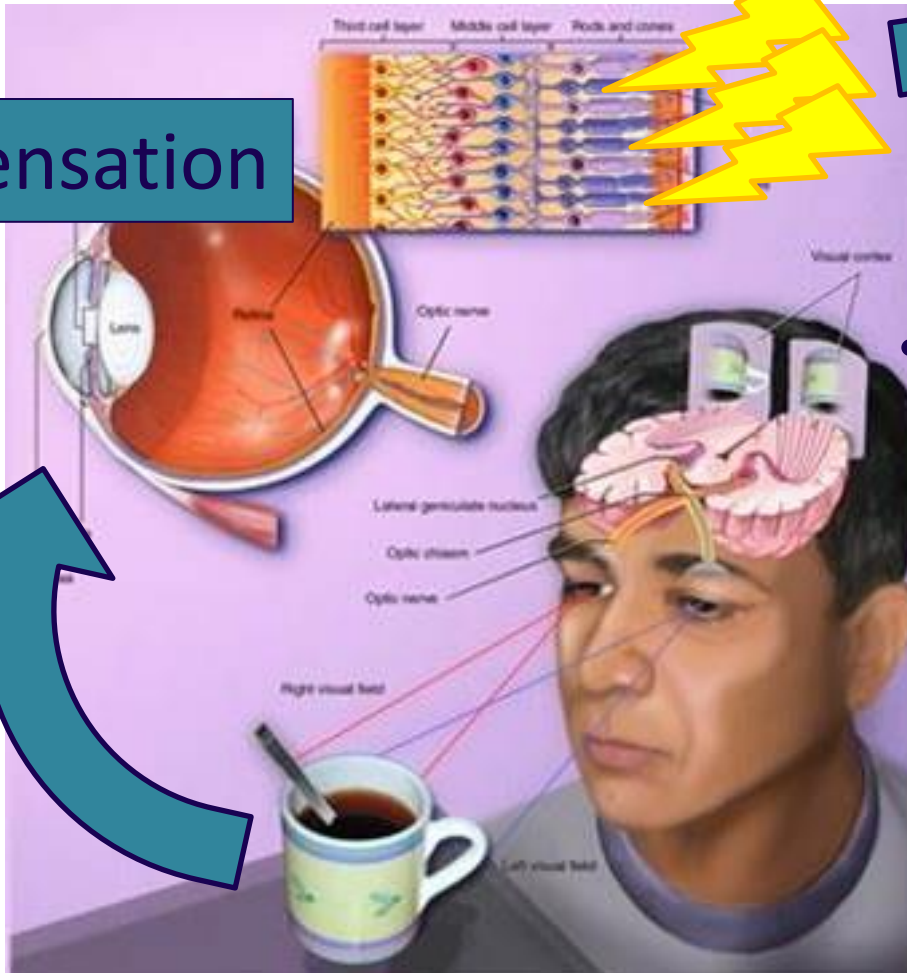
Cognition

Perception

Colour

Depth & size

Motion



Today...

Sensation

Cognition

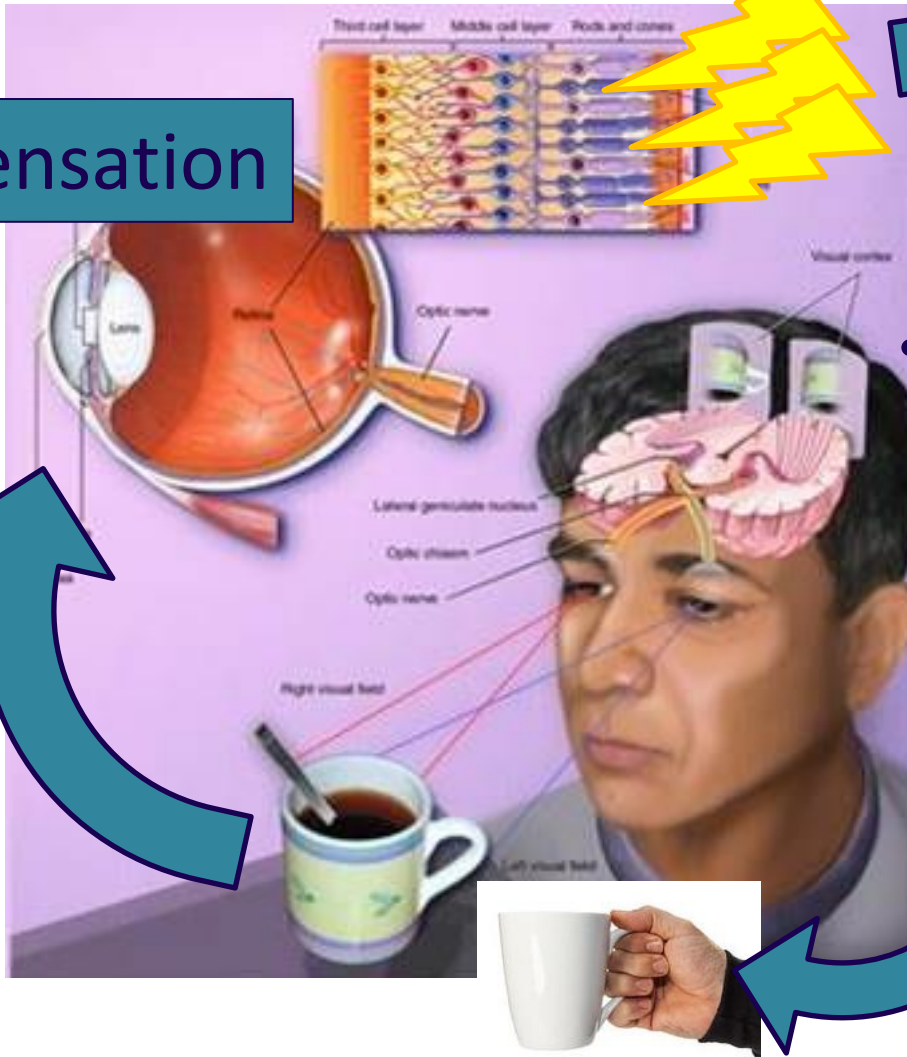
Perception

Depth & size

Action

Why is this important?

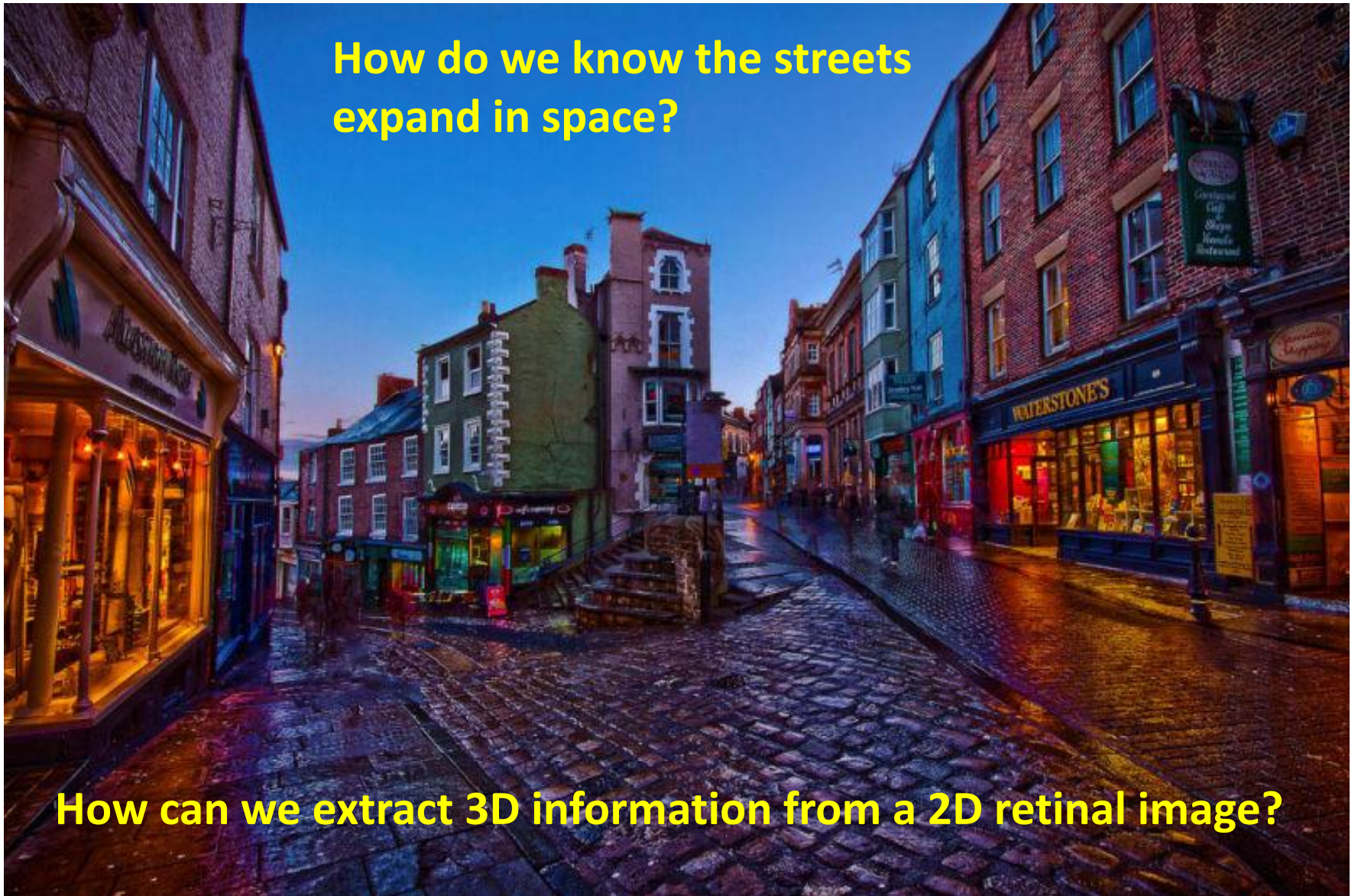
Depth and size are **veridical** object features.



Depth perception

How do we know the streets expand in space?

How can we extract 3D information from a 2D retinal image?

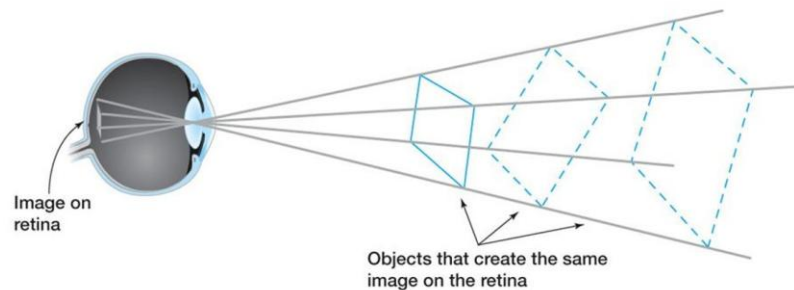


Depth perception

How do we know the streets expand in space?

- Ambiguous retinal images

– The inverse projection problem



How can we extract 3D information from a 2D retinal image?

Depth cues

- Body-related
 - **Oculomotor** cues (state of the eyes).
- Stimulus-related
 - **Monocular** cues (work with one eye).
 - **Binocular** cue (depends on both eyes).

Depth cues

We **feel** that an object is close to us or far away.

- Body-related
 - **Oculomotor** cues (state of the eyes).

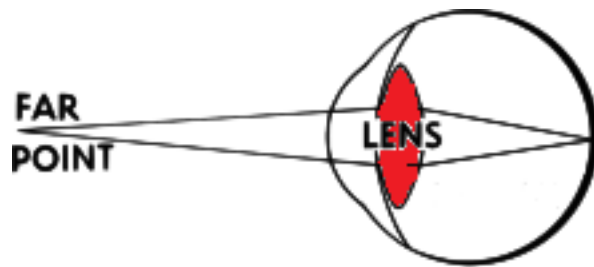
Depth cues: Oculomotor cues

- Oculomotor cues

Accommodation

Convergence

Relaxed ciliary muscles

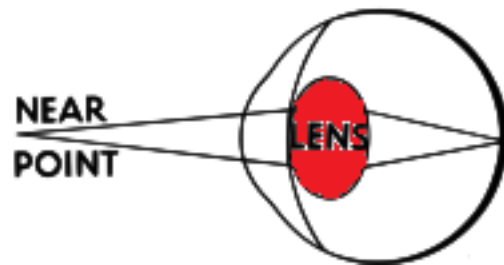
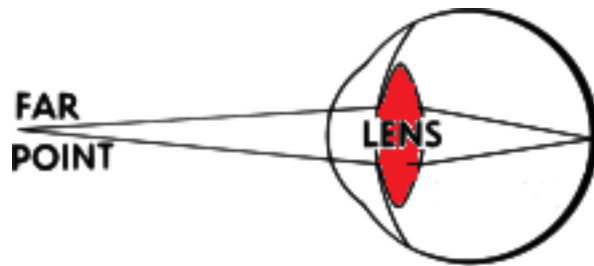


Depth cues: Oculomotor cues

- Oculomotor cues

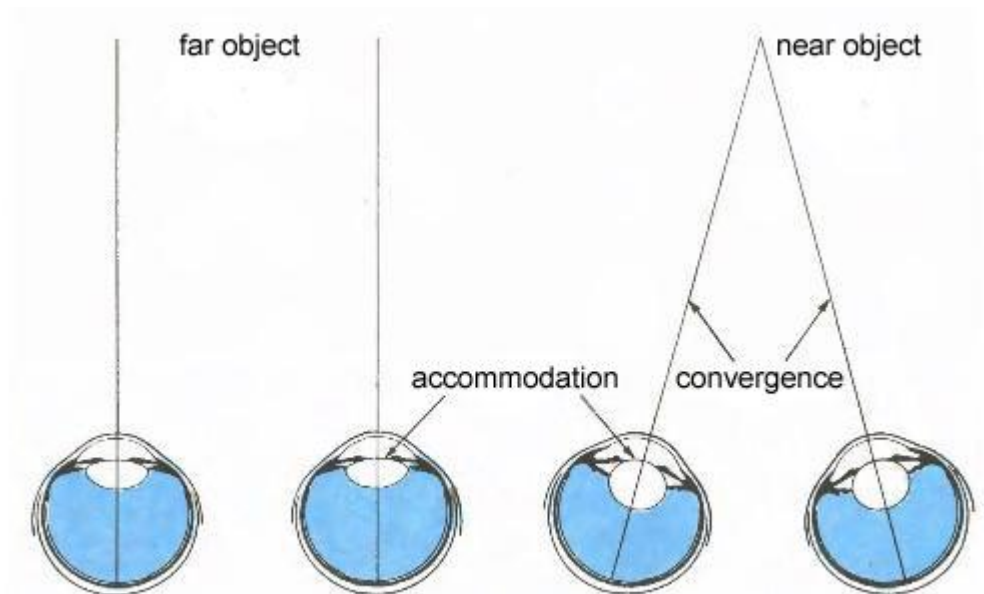
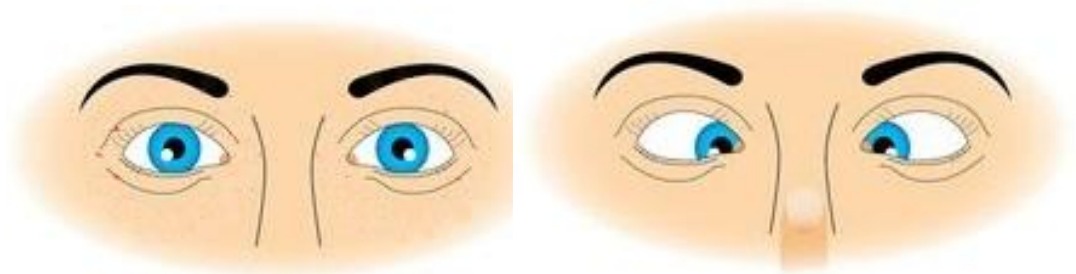
Accommodation

Relaxed ciliary muscles



Tightened ciliary muscles

Convergence



Depth cues

- Body-related
 - **Oculomotor** cues (state of the eyes).
Accommodation, Convergence
- Stimulus-related
 - **Monocular** cues (work with one eye).

Depth cues: Monocular cues

- **Pictorial cues**

- Occlusion
- Relative height
- Relative size
- Perspective convergence
- Familiar size
- Atmospheric perspective
- Texture gradient
- Shadows

We extract 3D information from still 2D images

Depth cues: Monocular cues → Pictorial cues

- Occlusion

Occluded (partially hidden) objects are further away than occluding objects.

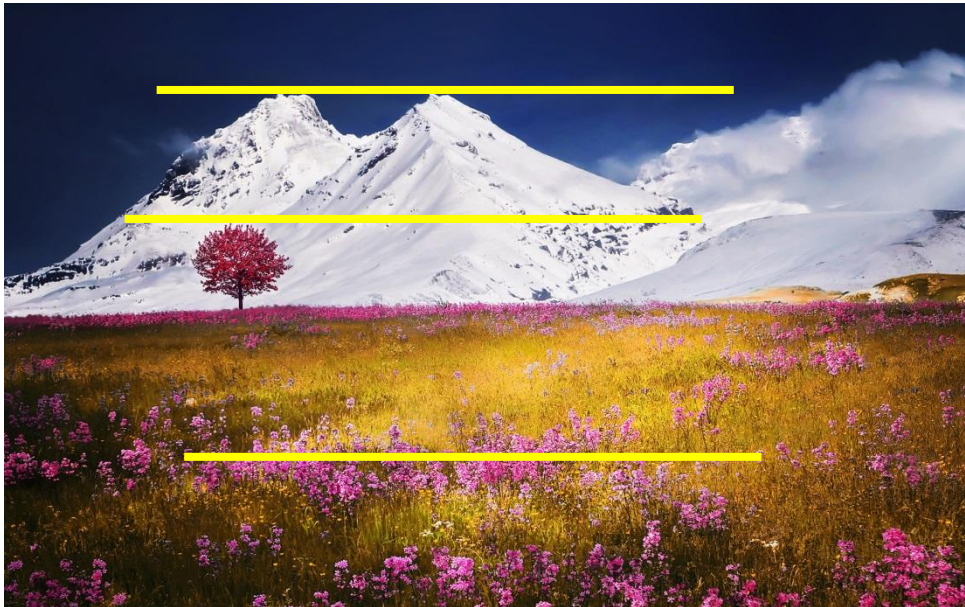


Depth cues: Monocular cues → Pictorial cues

- Relative height

Objects **higher in the visual field** are further away than objects lower in the visual field.

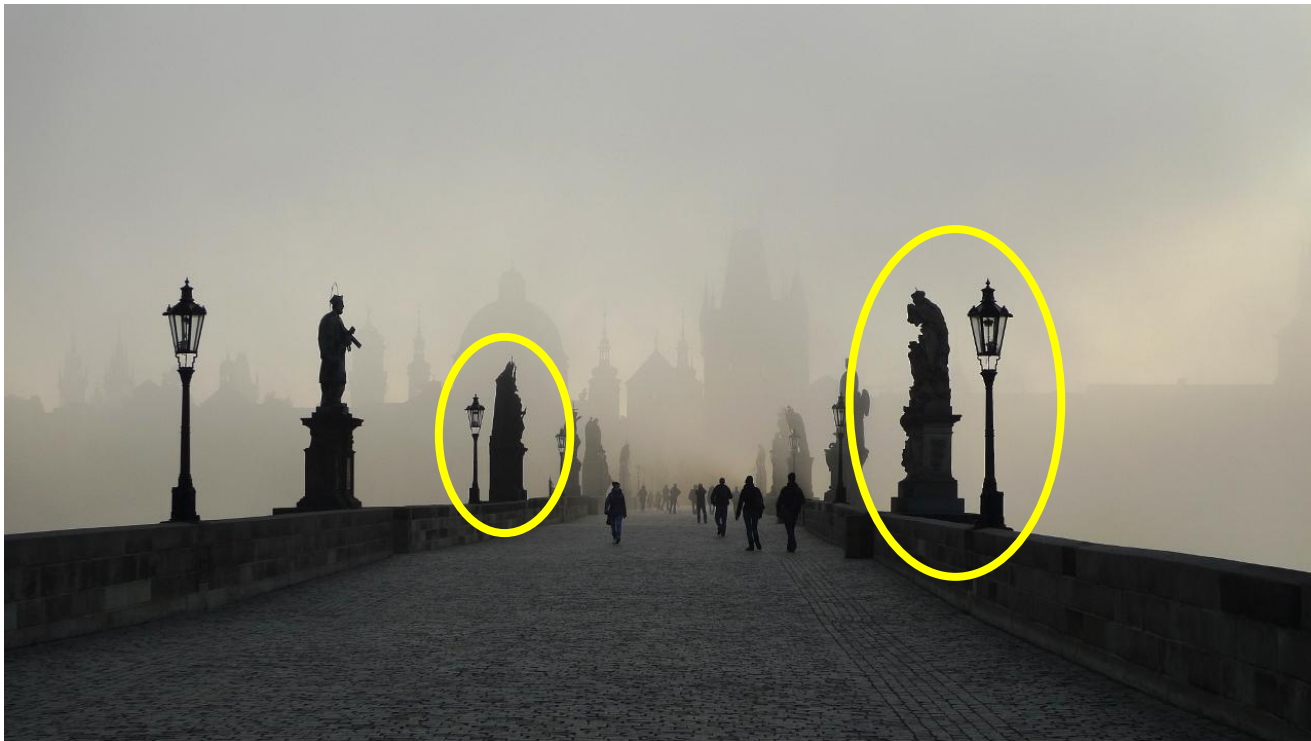
Unless we are on a hill, then objects with their **bases closer to the horizon** are further away.



Depth cues: Monocular cues → Pictorial cues

- Relative size

When two objects are equal in size, the one that is further away will take up **less of the visual field**.



Depth cues: Monocular cues → Pictorial cues

- Familiar size

Knowledge about physical size of objects.

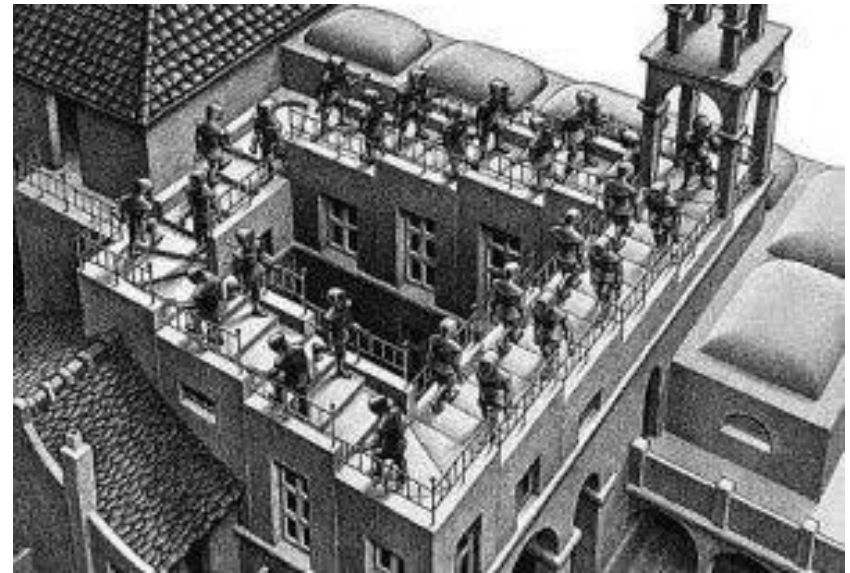


e.g., Epstein (1965) **experiment**:
One pound coin is judged to be further away, because it is known to be larger than the 5p coin.

Depth cues: Monocular cues → Pictorial cues

- Perspective convergence

Converging parts of objects are further away.



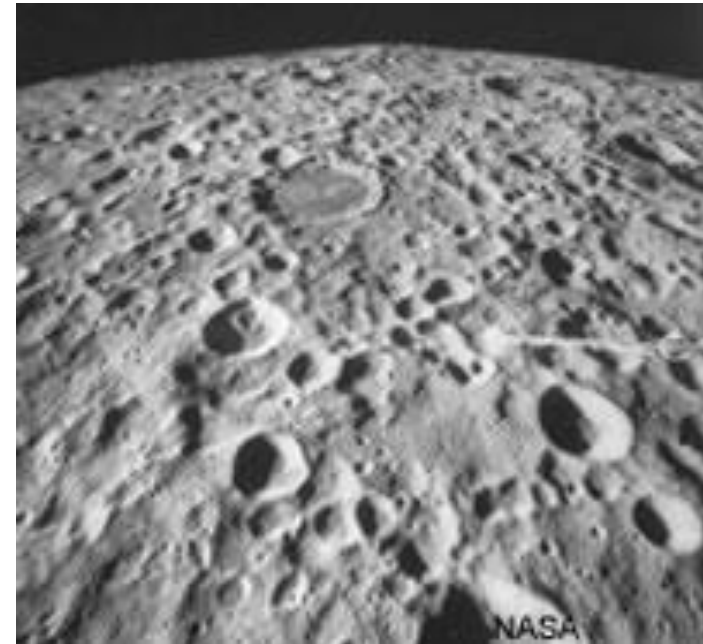
Painting by Escher

Depth cues: Monocular cues → Pictorial cues

- Atmospheric perspective

Objects further away are **less sharp** and have a **blue tint**.

Particles in the atmosphere scatter light (preferentially short wavelengths) which softens and colours further away objects.



Not on the moon...

Depth cues: Monocular cues → Pictorial cues

- Texture gradient

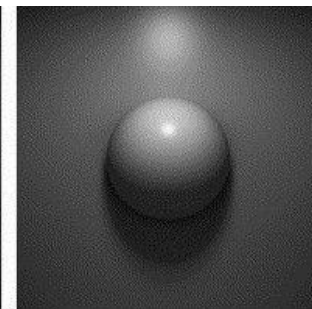
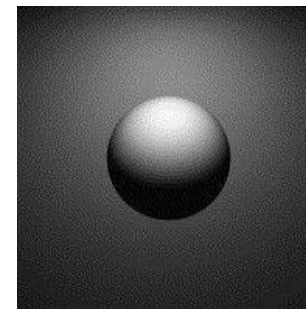
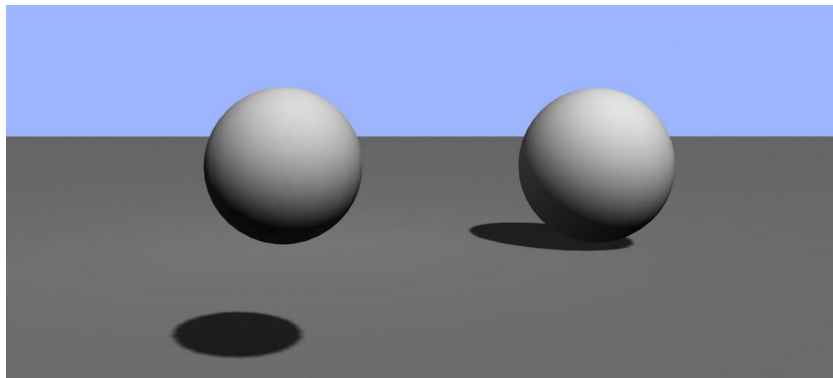
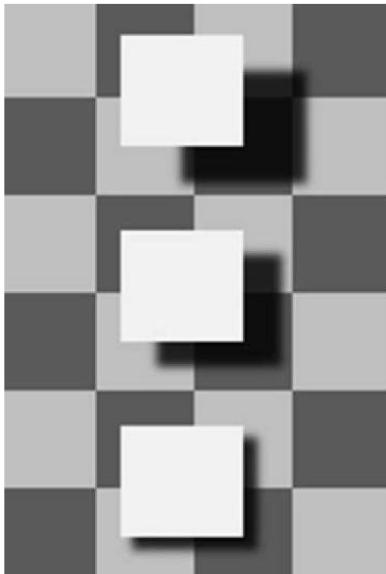
Denser textures indicate further away objects.



Remember Gibson's invariant visual information?

Depth cues: Monocular cues → Pictorial cues

- Shadows



Depth cues: Monocular cues

- **Pictorial cues**

- Occlusion
- Relative height
- Relative size
- Perspective convergence
- Familiar size
- Atmospheric perspective
- Texture gradient
- Shadows

We extract 3D information from still 2D images

- **Motion-based cues**

- Motion parallax
- Deletion & Accretion

We extract 3D information from animated 2D images

Depth cues: Monocular cues → Motion-based cues

- Motion parallax

Nearby objects **glide away quickly**,
far away objects **glide along slowly**.



Depth cues: Monocular cues → Motion-based cues

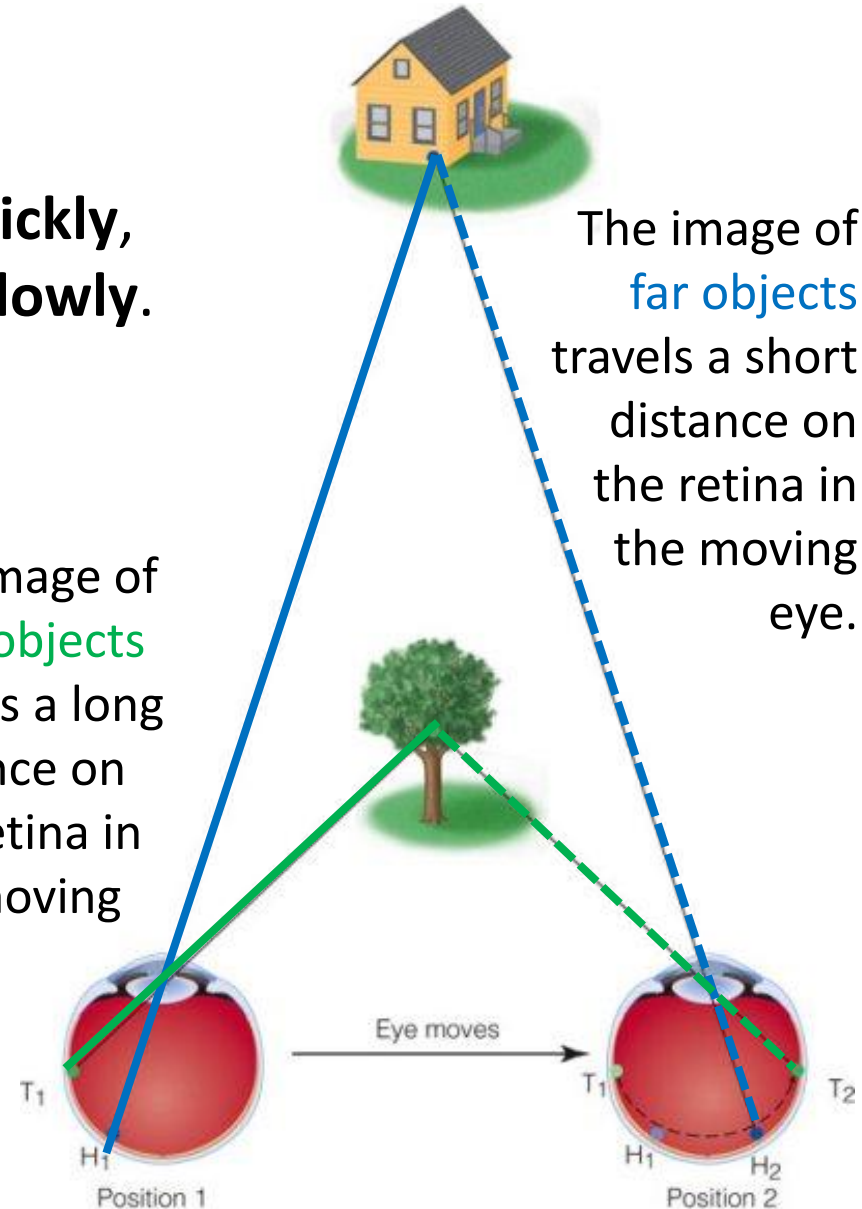
- Motion parallax

Nearby objects **glide away quickly**,
far away objects **glide along slowly**.



The image of **near objects** travels a long distance on the retina in the moving eye.

The image of **far objects** travels a short distance on the retina in the moving eye.



Depth cues: Monocular cues → Motion-based cues

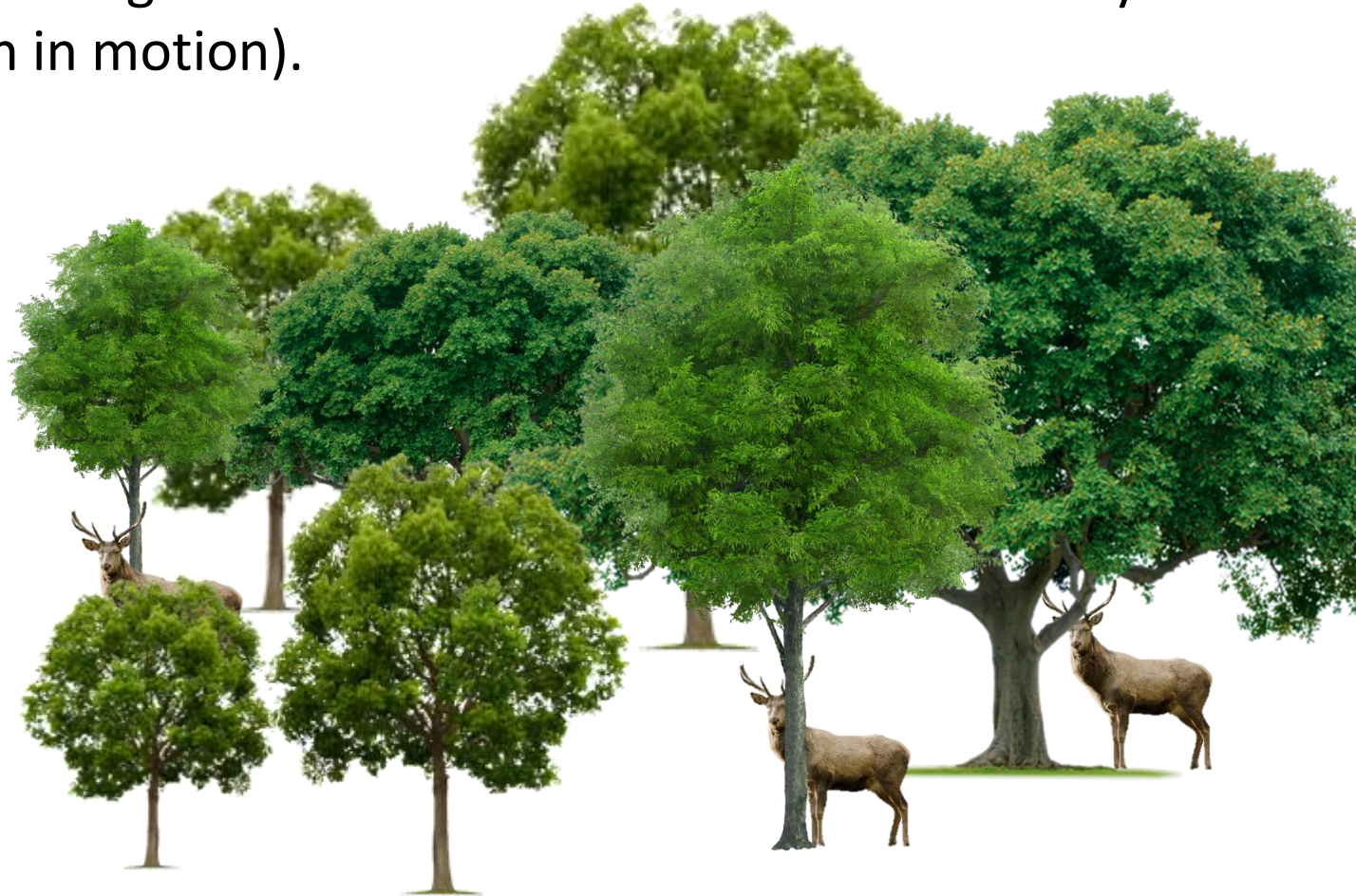
- Deletion & accretion

An object being **covered and uncovered** is further away (occlusion in motion).

Depth cues: Monocular cues → Motion-based cues

- Deletion & accretion

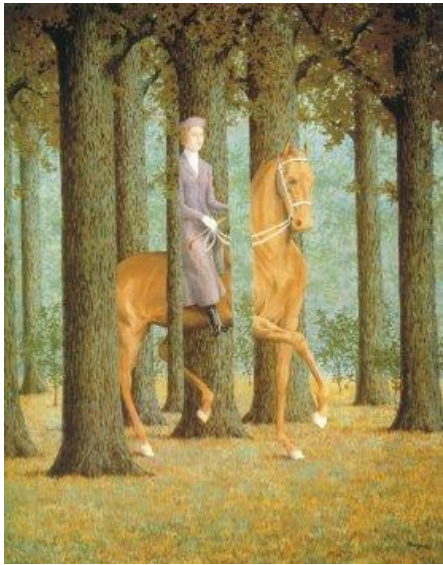
An object being **covered** and **uncovered** is further away (occlusion in motion).



Depth cues: Monocular cues → Motion-based cues

- Deletion & accretion

An object being **covered** and **uncovered** is further away (occlusion in motion).



Painting by Rene Magritte



Depth cues

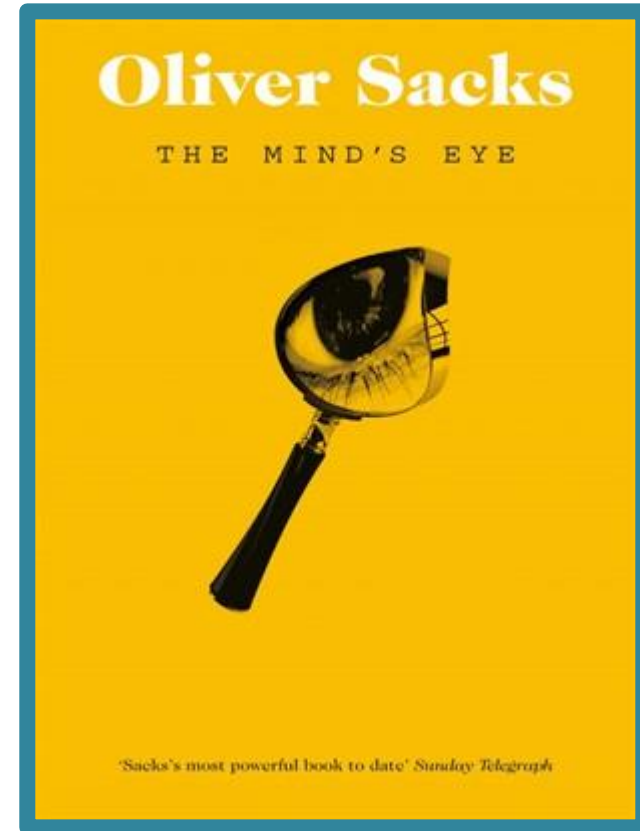
- Body-related
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 - Pictorial**: Occlusion, relative height, relative size, perspective convergence, familiar size, atmospheric perspective, texture gradient, shadows
 - Motion-based**: Motion parallax, deletion & accretion
 - **Binocular** cue (depends on both eyes).
Retinal disparity **The most powerful of all depth cues!**

Depth cues → Binocular cue

- Binocular vision
 - = **stereoscopic vision**
 - = **vision with both eyes**

The Stereoscopic Society was founded in 1893. They have their own journal, regular London meetings, an annual conference, their own library, etc. Oliver Sacks was a member...

<http://www.stereoscopicsociety.org.uk/WordPress/>

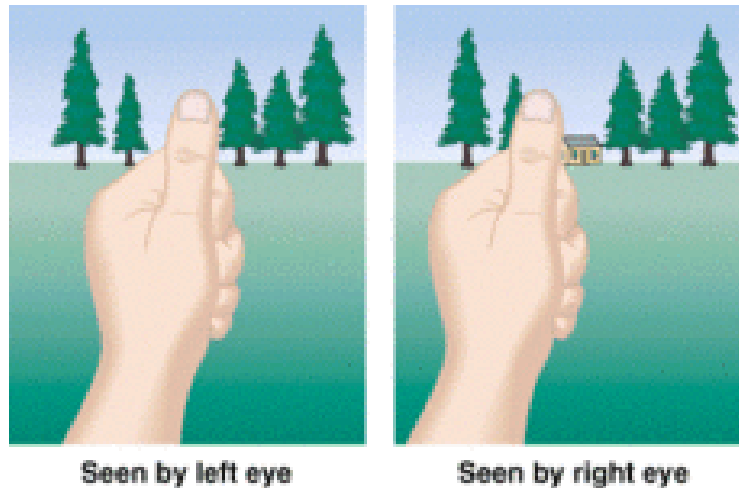


Look at Oliver Sacks and 'Stereo Sue' talking about how it feels to gain stereoscopic vision after half a life without it: https://www.youtube.com/watch?v=308-n1_tBUk

Depth cues → Binocular cue

- **Retinal disparity**

The retinal images in the left and right eye are slightly shifted (because our eyes are a few centimetres apart).

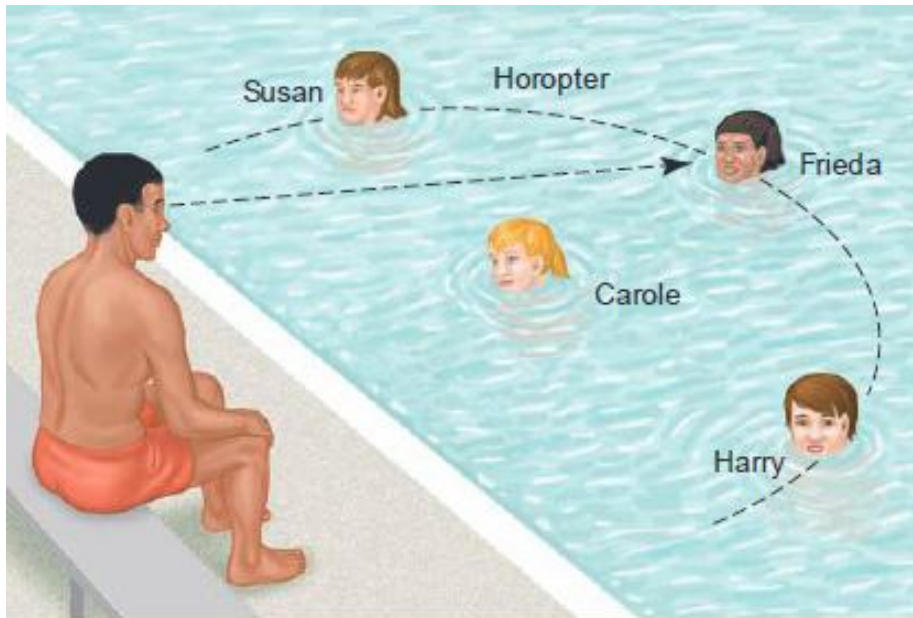


Note: The point of fixation is crucial for binocular depth perception.

Depth cues → Binocular cue

Corresponding retinal points

Dad fixates Frida, i.e., Frida's representation is in Dad's foveae (in both eyes).

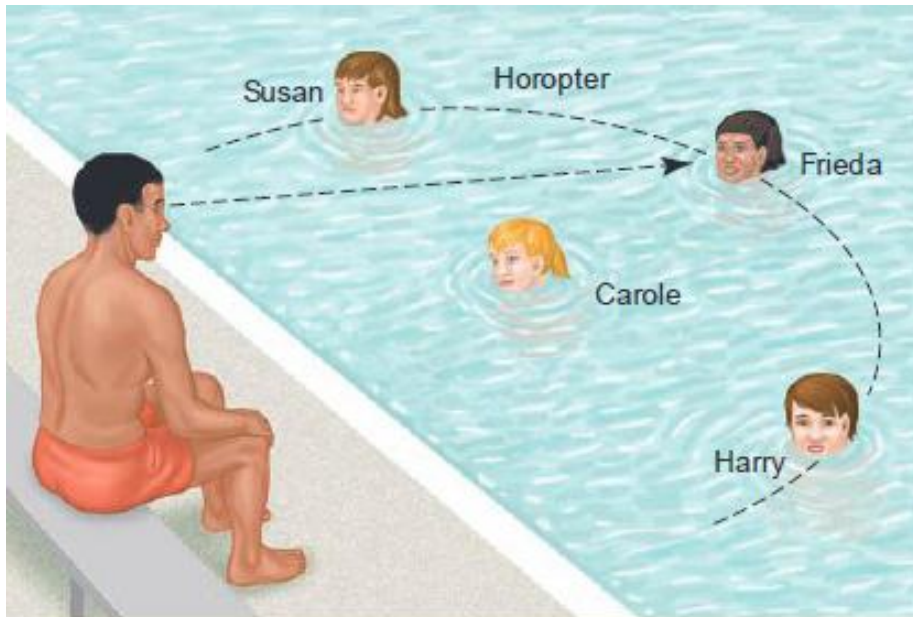
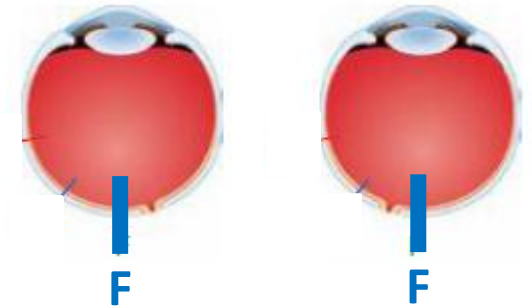


Depth cues → Binocular cue

Corresponding retinal points

The two foveae are corresponding points.

If eyes are overlaid
corresponding points
overlap exactly.



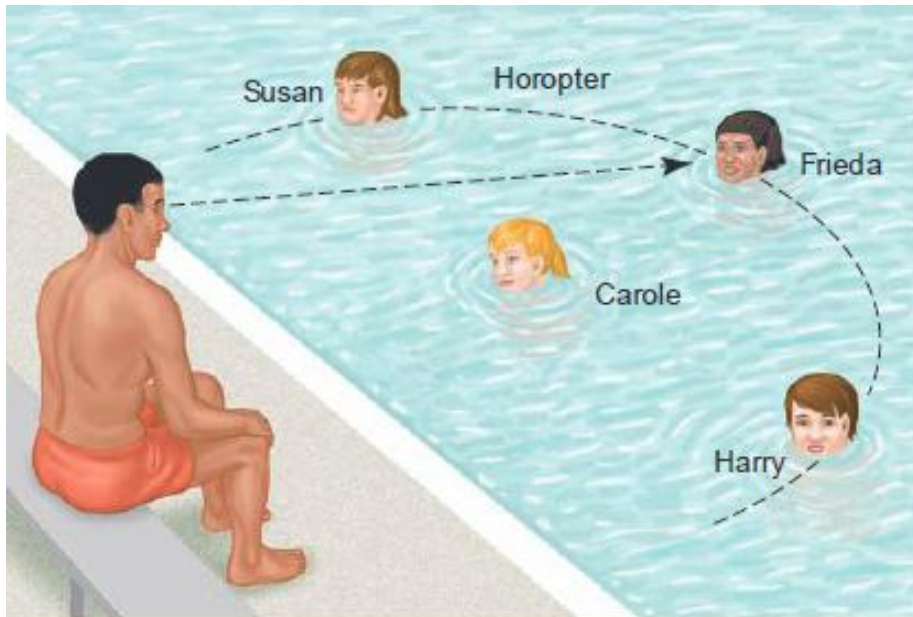
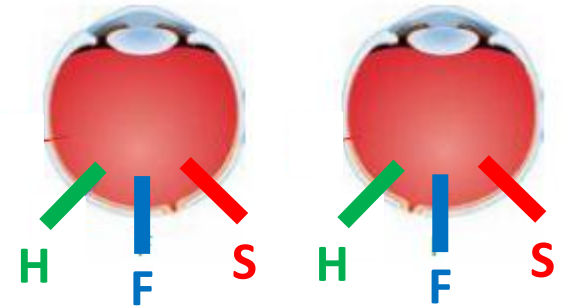
Depth cues → Binocular cue

Corresponding retinal points

The two foveae are corresponding points.

All objects with the **same distance as the fixation point** will be represented on corresponding retinal points (same relative position in both eyes).

If eyes are overlaid corresponding points overlap exactly.



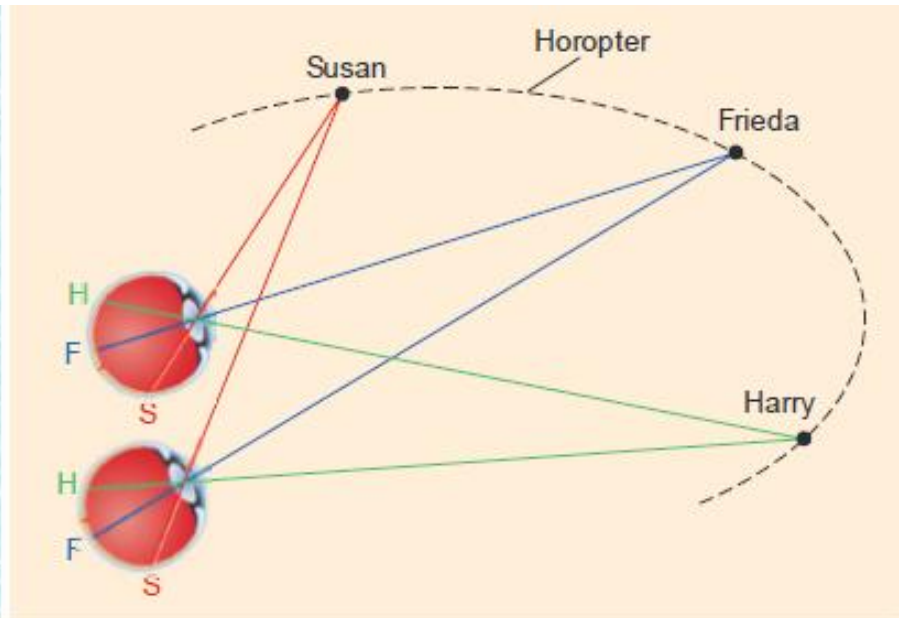
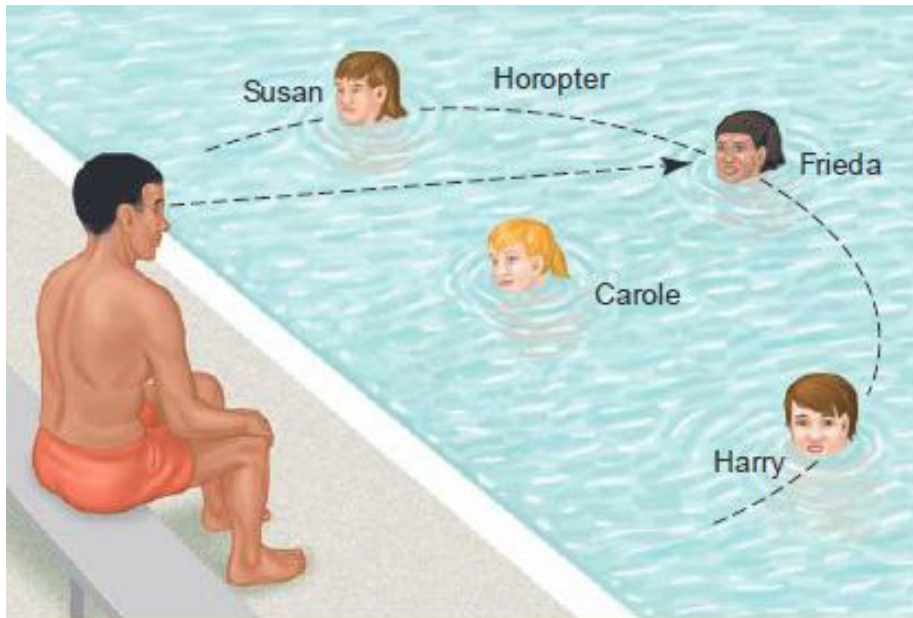
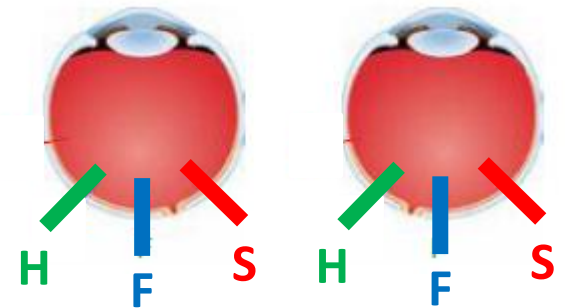
Depth cues → Binocular cue

Corresponding retinal points

Horopter: Imaginary plane through the fixation point connecting all objects for which retinal images fall on corresponding retinal points.

(Connects all objects that are **equidistant** from observer.)

If eyes are overlaid corresponding points overlap exactly.



Depth cues → Binocular cue

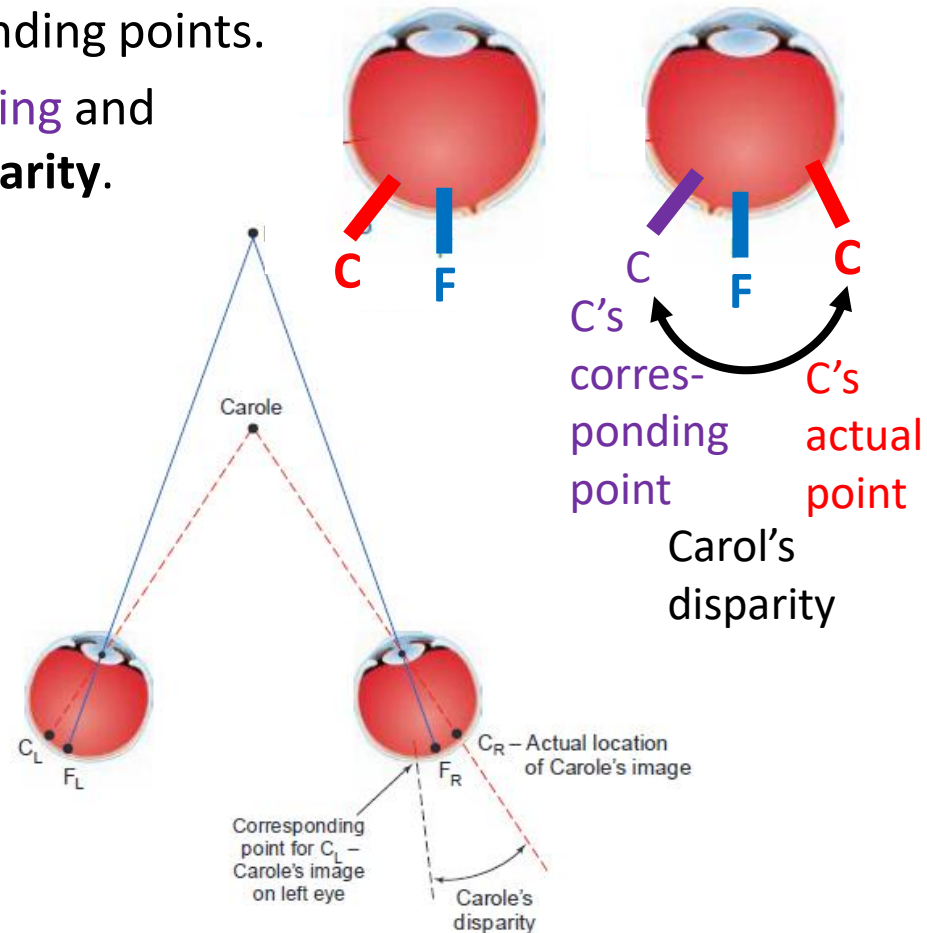
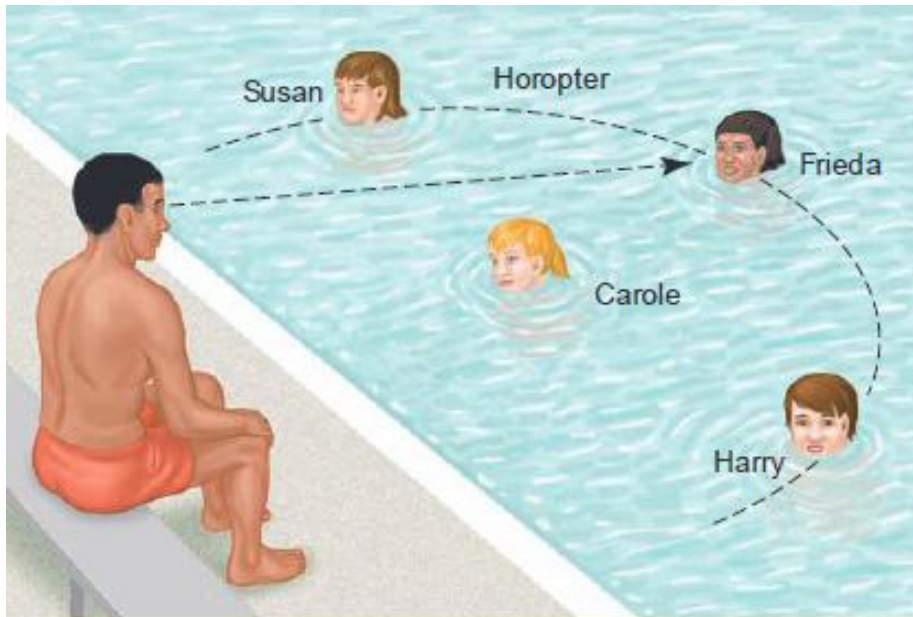
Non-corresponding retinal points

Carole is not on the horopter, she is closer to Dad than Frida.

Carole's retinal images have non-corresponding points.

The distance between Carole's **corresponding** and **actual** retinal points is Carole's **retinal disparity**.

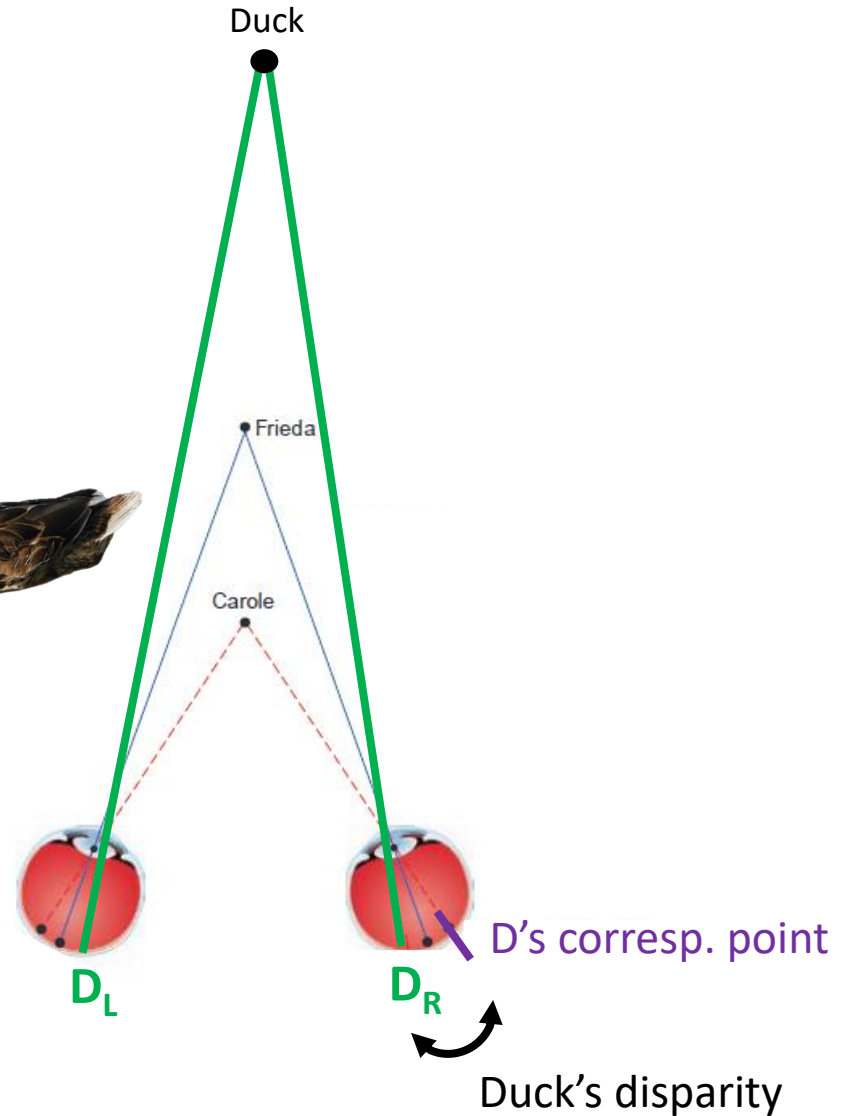
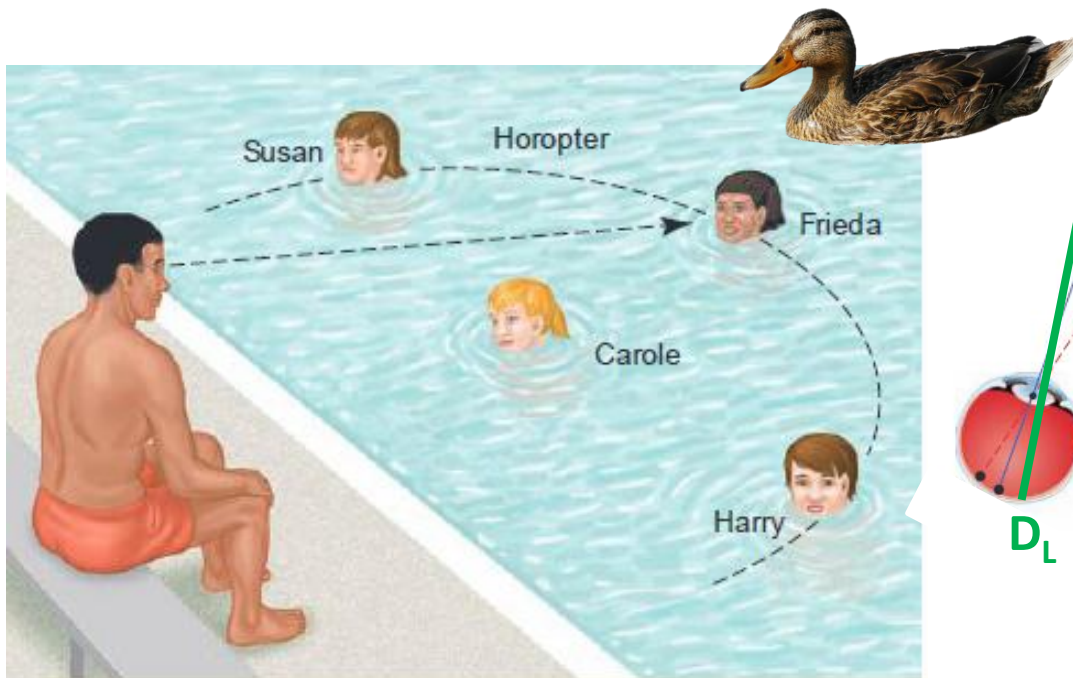
If eyes are overlaid
non-corresponding
points do not overlap.



Depth cues → Binocular cue

Non-corresponding retinal points

Same for objects that are further away than the point of fixation.



Depth cues → Binocular cue

- Stereopsis

- The ability to use **retinal disparity** as a **cue** to perceive depth.

Each distance from fixation produces a different amount of retinal disparity.

The larger the disparity, the further away is the object from fixation.

- Neuronal evidence

Disparity-selective neurons in V1 and along both ventral and dorsal streams.

Depth cues → Binocular cue

Is disparity sufficient to create depth perception - can disparity transform 2D images in 3D perception?

Depth cues → Binocular cue

Is disparity sufficient to create depth perception - can disparity transform 2D images in 3D perception?

Stereoscope/Stereograph



Holmes stereoscope (Oliver Holmes, 1861)



Depth cues → Binocular cue

Is **disparity sufficient** to create depth perception - can disparity transform 2D images in 3D perception? **Yes, looks like it...**



Depth cues → Binocular cue

- BUT
 - Stereoscopic photographs also contain **pictorial cues** which create depth perception.

Occlusion

Relative height

Perspective convergence

Atmospheric perspective

Texture gradient



Depth cues → Binocular cue

- Bela Julesz, 1971
 - Random dot stereogram



Yes, retinal disparity can transform 2D images into 3D perception!



If you don't believe it, get an auto-stereogram in which you can see 3D vision generated by retinal disparity without the help of any other depth cues.

Depth perception - summary

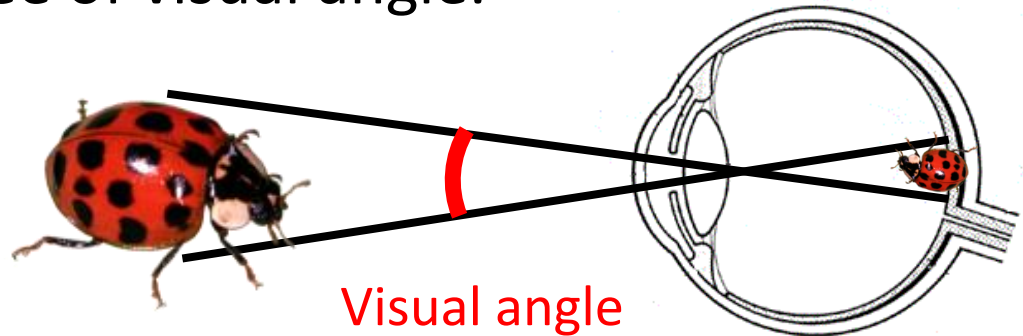
- How do we perceive depth? With the help of cues...
 - **Body-related cues**
 - Oculomotor** cues (state of the eyes): Accommodation, convergence
 - **Stimulus-related cues**
 - We identify information on the retinal image that is correlated with depth in the visual field.
 - Monocular** cues (work with one eye):
 - Pictorial**: Occlusion, relative height, relative size, perspective convergence, familiar size, atmospheric perspective, texture gradient, shadows
 - Motion-based**: Motion parallax, deletion & accretion
 - Binocular** cue (depends on both eyes): Retinal disparity

Size perception



Size perception

- Size of retinal object representations
 - Measured in degree of visual angle.

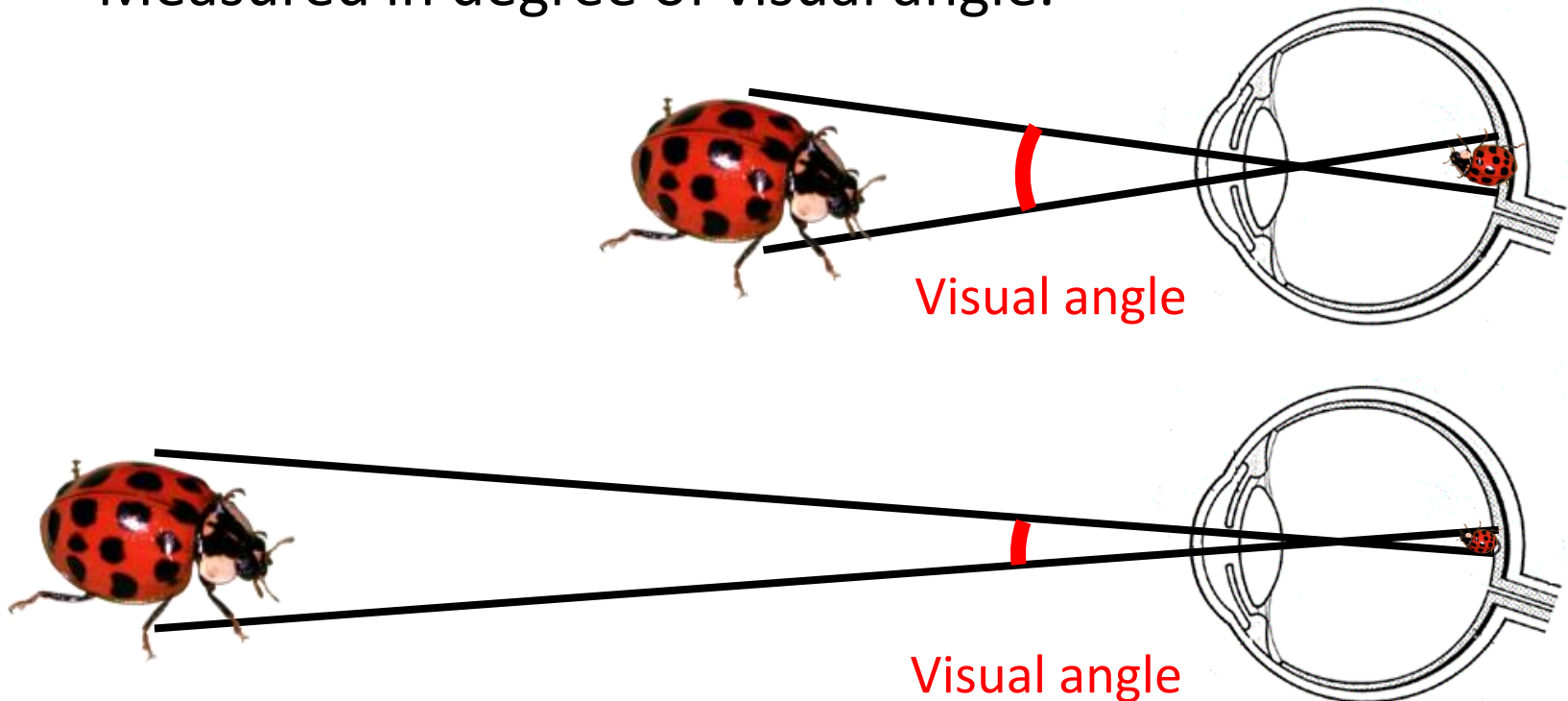


Depends on **physical object size** and **physical distance** from the observer.

If you want to know how exactly visual angles are calculated and why, look at this page:
<http://elvers.us/perception/visualAngle/>

Size perception

- Size of retinal object representations
 - Measured in degree of visual angle.



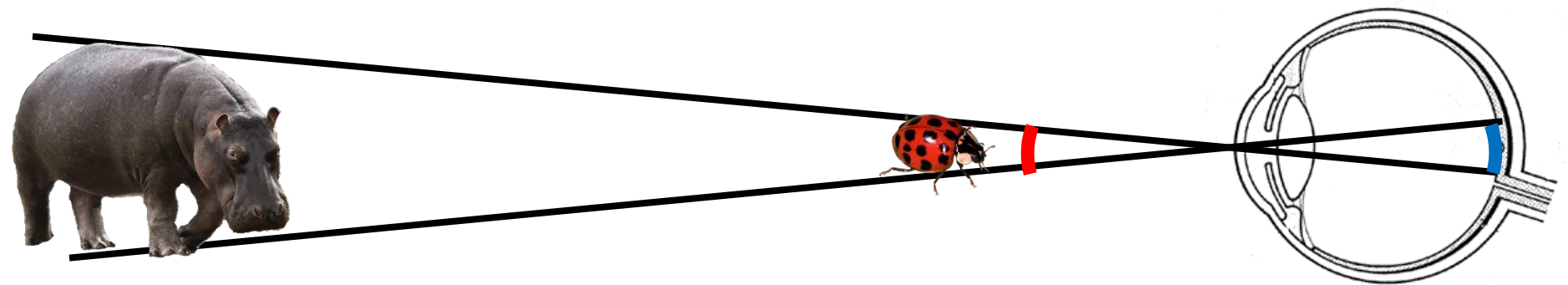
The same object has a larger visual angle when it is closer to the observer and a smaller visual angle when it is further away.

Size perception

- Size of retinal object representations
 - Measured in degree of visual angle.

The angle of an object relative to the observer's eye.

The size of the retinal object representation.



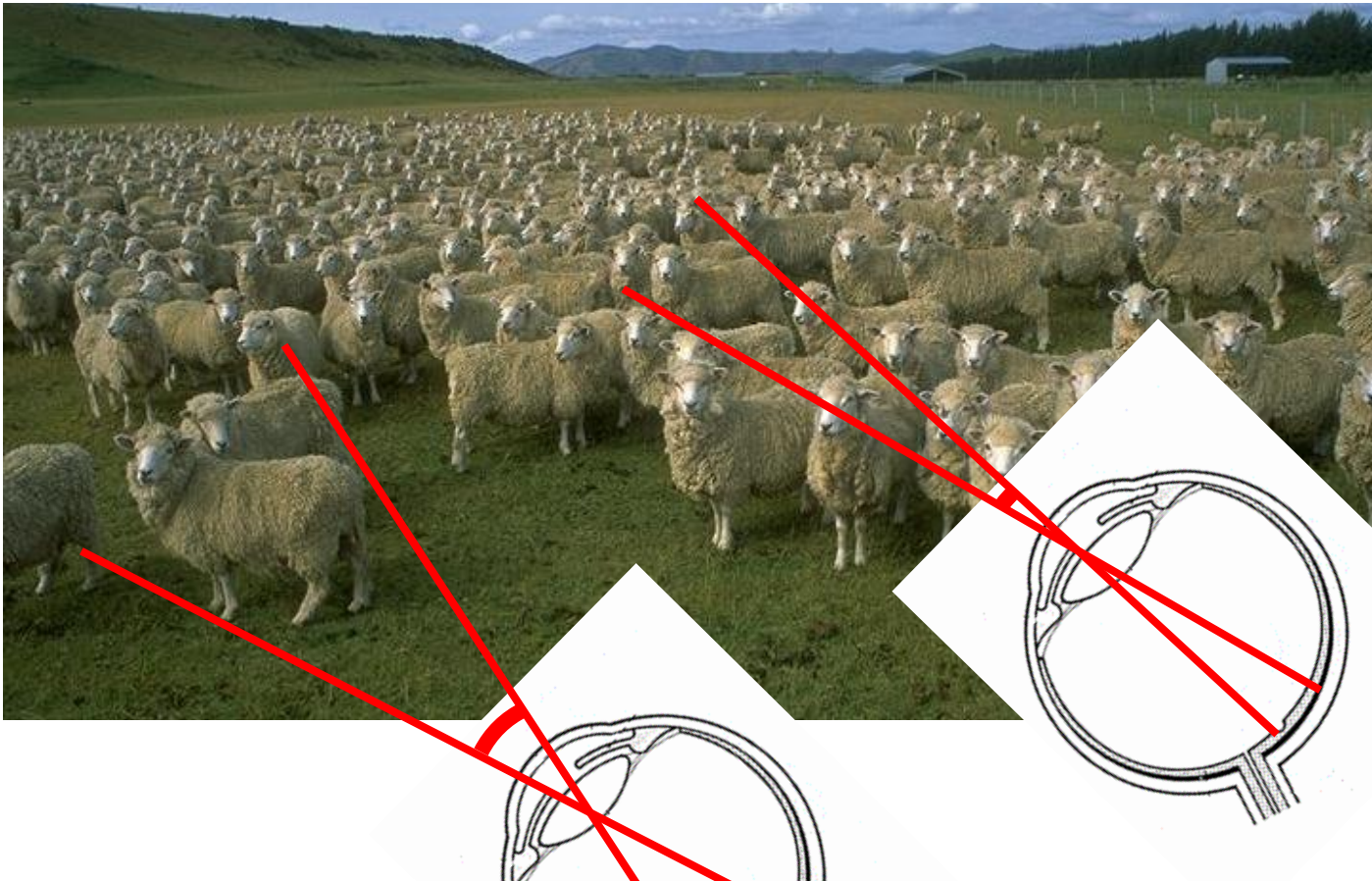
Two differently sized objects can have **the same visual angle** when the larger one is further away.



Size perception

- Size constancy

Perception of an object's size is relatively **constant** even when viewed from different distances (producing different visual angles).



Size perception

- Calculating size constancy
 - **Size-distance scaling equation**

Far away sheep: 2 = K(1x2)

Near sheep: 2 = K(2x1)

$$S = K(R \times D)$$

S	R	D
Perceived size	Size of the retinal image	Perceived distance

If a sheep walks towards you, the size of the sheep's image on your retina (R) gets larger, but your perception of the sheep's distance (D) gets smaller.

The perceived size (S) of the sheep remains the same.



Size perception

Size-distance scaling equation explains relative size illusions

$$S = K(R \times D)$$

Perceived size Retinal image size Perceived distance

Dino: $2 = K(2 \times 1)$

People: $1 = K(1 \times 1)$



The retinal image size is larger for the dino than the people.

Lack of pictorial depth cues:
We assume that the distance of the dino and the people is the same.

Hence

The size of the dino is perceived to be bigger than the size of the people.

Size perception

- Size-distance scaling equation explains the Ponzo illusion.

Rabbit 1: 2 = K(1x2)

Rabbit 2: 1 = K(1x1)

$$S = K(R \times D)$$

Perceived size Retinal image size Perceived distance



Rabbit 1

Rabbit 2



R: Rabbit 1 = Rabbit 2.

D: Rabbit 1 is perceived to be further away (pictorial depth cue).



Hence

S: Rabbit 1 is perceived to be larger.

Size perception

- Size-distance scaling equation explains the moon illusion.



Moon 2



Moon 1

$$S = K(R \times D)$$

Perceived size Retinal image size Perceived distance

R: Moon 1 = Moon 2.

D: Moon 1 is perceived to be further away (pictorial depth cues).

Hence

S: Moon 1 is perceived to be larger.

Size perception

- Same explanation for the Ponzo and moon illusion.

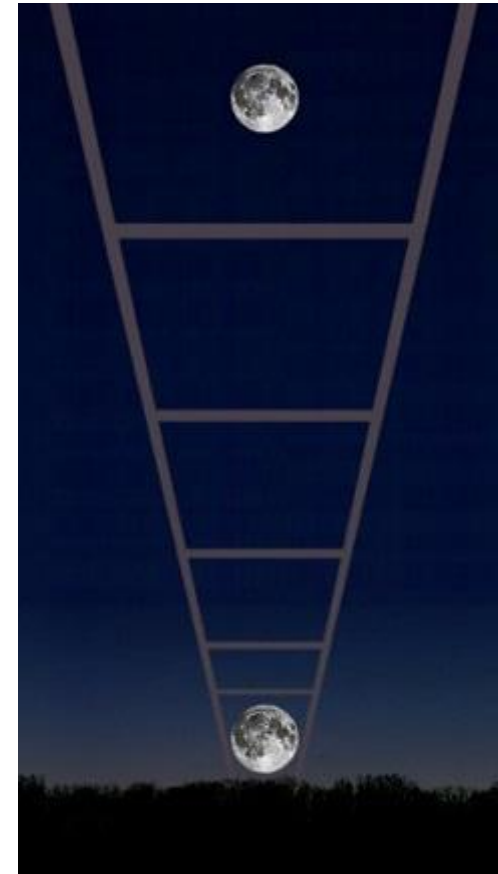
Rabbit 1: $2 = K(1 \times 2)$

Rabbit 2: $1 = K(1 \times 1)$



Moon 1: $2 = K(1 \times 2)$

Moon 2: $1 = K(1 \times 1)$



Moon 2

Moon 1

Summary

- We have looked at
 - Cue-based depth perception.

Body-related (**oculomotor**) cues and **monocular** (pictorial, motion-based) and **binocular** (retinal disparity) stimulus-related cues.

Retinal **disparity** (shifted retinal images in the two eyes) as the most powerful depth cue.

 - Leads to **stereopsis** (is sufficient to create depth perception).
 - Size perception

Measurements of retinal image size (degree of visual angle).

Size-distance scaling (size constancy).

Test yourself – learning goals

- After reviewing this lecture, you should be able to
 - Name 3 different categories of depth cues and give examples.
 - Explain how depth cues (accommodation, convergence, occlusion, relative height, relative size, perspective convergence, familiar size, atmospheric perspective, texture gradient, shadows, motion parallax, deletion & accretion, and retinal disparity) shape perception.
 - Define retinal disparity and stereopsis and explain why we know that retinal disparity is sufficient to elicit stereopsis.
 - Discuss the factors that modulate visual angle.
 - Explain size constancy and the factors used to calculate it.

Test yourself – multiple choice

1. Depth perception is based on a number of cues. Which of the following cue pairs is not 100% related to depth perception?

- a) Accommodation and convergence.
- b) Relative height and size.
- c) Deletion and accretion.
- d) Texture gradient and gravity.

Test yourself – multiple choice

2. Which of the following rules derived from depth cues is not correct?

- a) Far accommodated objects are further away.
- b) Occluding objects are further away.
- c) Blurry objects are further away.
- d) Objects with denser texture are further away.

Test yourself – multiple choice

3. What is not described by visual angle?

- a) The angle of an object with respect to the observer's eye.
- b) The physical size of an object.
- c) The size of the retinal object representation.
- d) The angular degree an object image occupies on the retina.

See you next week!

